

# Georgia Dimaki

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**Summary:** A versatile software engineer who combines hands-on experience in systems programming and pipeline architecture with the rigorous, analytical thinking from an MIT Operations Research background. This unique blend of practical engineering and theoretical modeling is applied to building robust, performant, and adaptable software systems.

## EDUCATION

**Massachusetts Institute of Technology**  
*Master of Science in Operations Research*

Aug. 2018 – Sept. 2020  
Cambridge, MA

- **GPA:** 4.6/5.0

**Athens University of Economic and Business**  
*Bachelor of Science in Computer Science*

Oct. 2013 – Apr. 2018  
Athens, Greece

- **GPA:** 9.69/10.0 | **Honors:** Ranked 1st in Graduating Class, Multiple Awards for Academic Excellence.

## PROFESSIONAL EXPERIENCE

**InterSystems**  
*Senior Systems Engineer*

Nov. 2020 – Present  
Cambridge, MA

- Architected and implemented a high-performance, cross-platform **C++ dynamic library** to unify multi-language API support for the company's core data platform, **reducing the development cycle for new language APIs from 6 months to 1 month**.
- Developed a performant **Node.js API** for client-server communication, leveraging the **N-API** to create efficient bindings to the underlying C++ library.
- Mentored new team members on system architecture and best practices, improving team onboarding and overall productivity.
- Redesigned a human-written service onboarding manual into an **AI-navigable platform**, comprising a multi-skill plugin (manual, automation-aware, and expert-level skills), a **Python automation library**, and a VS Code extension providing a virtual view of service implementation files.
- Implemented a **LangChain-compatible vector store driver** for InterSystems IRIS, enabling RAG pipelines to leverage the platform's native vector capabilities as a drop-in replacement for standard retrieval backends.
- Built a **proof-of-concept MCP server** to explore agentic tool-use patterns, transforming a conventional UI into a conversational agent experience for internal developer tooling.
- Designed and delivered an internal training series on **RAG architectures and MCP**, developing hands-on Jupyter notebooks to demonstrate LLM integration patterns and help colleagues reason about AI-assisted workflow automation.

**Laboratory for Information and Decision Systems (LIDS), MIT**  
*Graduate Researcher; Advisor: Prof. Eytan Modiano*

Aug. 2018 – Jun. 2020  
Cambridge, MA

- Engineered a Python-based simulation environment to model and optimize network node clustering in large-scale data centers (up to 4096 ToR switches).
- Implemented and benchmarked iterative algorithms using **stochastic optimization** and **spectral clustering** to achieve rapid network reconfiguration in tens of milliseconds.

## PROJECTS

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### Agentic Angular App Generation | *Python, LLMs, Agent Orchestration*

- Built a skill for one-shot Angular application generation via implicit agent orchestration during a hackathon, coordinating multiple specialized agents to handle code generation subtasks end-to-end.

### Notebook Slide Preview VS Code Extension | *TypeScript, VS Code API, JavaScript*

- Developed a VS Code extension to instantly preview and present Jupyter/IPython notebooks as interactive slides directly within the editor.

### Distributed Control Platform for Autonomous Vehicles | *JavaScript, Kafka, Raspbian OS*

- Engineered a real-time, distributed platform to manage network protocols on Unmanned Surface Vehicles (USVs) using JavaScript and a Kafka messaging queue.
- Developed shell scripts to automate the deployment and configuration process across dozens of devices, ensuring consistent updates for the entire fleet.

### Computer Vision Pipeline for Gene Expression Prediction | *Python, PyTorch, Weights & Biases*

- Independent research exploring ML frameworks and deep learning approaches through an end-to-end computer vision pipeline to predict gene expression levels from high-resolution H&E stained histology images.
- Implemented custom data loaders for efficient processing of large image tiles and modified deep neural network (DNN) architectures to improve predictive accuracy.

### Robust Optimization for Ventilator Allocation | *Python, Gurobi, pandas*

- Developed a robust optimization model that improved ventilator availability by over **30%** compared to existing mixed-integer programming solutions during resource shortages.

### Hotel Cancellation Prediction Model | *R, XGBoost, caret*

- Engineered a predictive model using XGBoost that achieved **86% accuracy** in forecasting hotel cancellations, maintaining a false positive rate below 7%.

## SKILLS

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**Languages:** Python, C++, Node.js, JavaScript, Java, R, SQL, Julia, HTML/CSS

**Frameworks/Libraries:** React, Express.js, PyTorch, Tensorflow, scikit-learn, pandas, NumPy, LangChain

**LLM & Agent Tools:** MCP (Model Context Protocol), RAG, Vector Stores, LLM API integration

**Databases & Platforms:** InterSystems IRIS, MongoDB, Kafka

**Optimization Tools:** Gurobi, Julia/juMP, GPyOpt, AMPL

**Developer Tools:** Git, Github, Docker, Jira, Trello, Weights & Biases

## LANGUAGES & INTERESTS

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**Languages:** Greek (Native), English (Fluent), Russian (Beginner)

**Interests:** Rowing, dancing, singing, songwriting, clothes design